GridWise

Smarter power planning. Faster service for every community.

An AI Solution for Industries | Energy sector

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THE PROBLEM

- Municipal electricity utilities often plan supply with **averages and manual estimates**.
- Under-forecast → **overloads and outages**. Over-forecast → **wasted budget**.
- Fault and billing complaints queue up on phone lines and at counters.
- Result: unreliable supply, higher costs, frustrated communities.

OUR SOLUTION

FORECAST
LSTM + ML predict hourly demand 24 hours ahead

UNDERSTAND
NLP sorts fault reports by type and urgency

RESPOND
Chatbot answers outage, billing and fault queries

ONE DASHBOARD FOR THE UTILITY

BENEFITS

- **Fewer unplanned outages** - plan purchases and maintenance with confidence.
- **Lower operating costs** - buy only the power you need.
- **Faster 24/7 service** - automatic fault triage and a chatbot for residents.
- **Safer communities** - dangerous cable faults flagged as HIGH urgency.

1 DATA → **2 CLEAN** → **3 TIME SERIES ANALYSIS** → **4 TRAIN MODELS** → **5 EVALUATE** → **6 DEPLOY DASHBOARD**

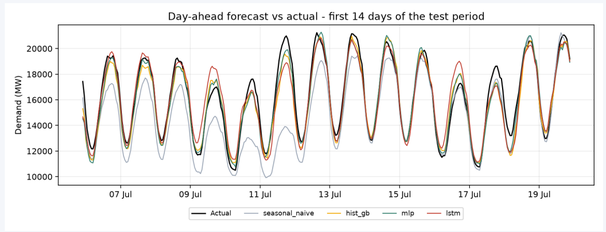
AI TECHNIQUES & DATA

Time series: decomposition, seasonality, autocorrelation
Machine learning: gradient boosting, MLP, SARIMA benchmark
Deep learning: LSTM on the previous 168 hours
NLP: TF-IDF + logistic regression fault triage
Chatbot: intent classifier + ticketing + live forecast

Data
- PJM Hourly Energy Consumption (hourly MW, 2002-2018)
- Synthetic fault reports & chatbot intents (labelled as synthetic)

Evaluation: chronological train/test split, walk-forward CV; MAE, RMSE, MAPE

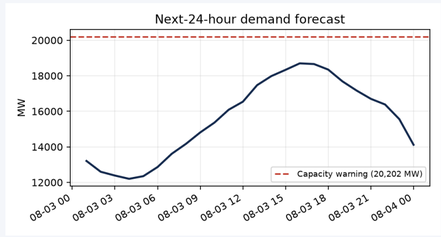
RESULTS (TEST PERIOD, REAL DATA)



Model	MAE (MW)	RMSE (MW)	MAPE
Seasonal naive	1445	1910	9.58%
Gradient boosting	697	919	4.67%
Neural net (MLP)	**689**	**911**	**4.61%**
LSTM	787	1029	5.35%

Fault classifier accuracy: 100% (synthetic reports - optimistic)
Chatbot intent accuracy (cross-validated): 81%

LIVE DEMO



User: there is a power line down outside my house sparking
GridWise: SAFETY FIRST: stay at least 10 metres away from fallen or damaged lines and keep other people and animals away. Do not touch anything in contact with the line. I've ...

User: what is the demand forecast for tomorrow
GridWise: Over the next 24 hours demand is forecast to peak at about 18,699 MW around 16:00. No hours are forecast above the capacity warning level (20,202 MW).

Limitations & future work: proof of concept trained on US data - a real deployment retrains on the municipality's own meter data. Next: weather inputs, smart-meter integration, local-language chatbot.

Sources: PJM Hourly Energy Consumption (Kaggle, robikscube); Bitext customer-support dataset (Hugging Face); scikit-learn. **Code:** github.com/BA-Practical-Assignment/GridWise.git

GitHub Repo